Organ-specific microplastic accumulation in channel catfish (*Ictalurus punctatus*) and bluegill (*Lepomis macrochirus*) in the Ohio River: Influence of habitat use and feeding ecology

Lindsay Barnes and **Chris Lorentz PhD.**
The Center for Ohio River Research & Education
Thomas More University Biology Field Station
8309 Mary Ingles Highway, California, KY 41007

ABSTRACT

Microplastic pollution poses a growing threat to freshwater ecosystems, yet species and organ-specific patterns of microplastic accumulation remain poorly understood. This study investigated the distribution and abundance of microplastics in three organs (gills, stomach, and swim bladder) of two common Ohio River fish species with contrasting habitat use and feeding ecologies: benthic-foraging channel catfish (*Ictalurus punctatus*) and pelagic-feeding bluegill (*Lepomis macrochirus*). Specimens were collected opportunistically. Each organ was digested, filtered, and analyzed under microscopy for particle counts. Microplastics were quantified as total particles and compared to organ mass. The mean total particle counts per organ (n=2) showed channel catfish exhibited the highest accumulation in stomach (6.5 microplastics) and swim bladder (6 microplastics), while the bluegill specimen showcased the same amount of microplastics as channel catfish (3.5 microplastics) in the gills. These organ-specific differences reflect habitat and feeding strategy in freshwater systems.

INTRODUCTION

Microplastics are defined as plastic particles smaller than 5mm in diameter (GESAMP, 2015, 2016). This definition of microplastics has been reformulated and investigated in light of its recent rise in the scientific community worldwide. They can be distinguished according to their origin, specifically primary microplastics, which are plastics produced to be of that size, such as facial cleanse exfoliation beads or seed coatings. Alternatively, they can be secondary microplastics, which are plastics that have undergone degradation processes in the environment, thereby polluting it.

These pollutants can enter aquatic environments through various means, including wastewater treatment plants, terrestrial runoff, and improper waste disposal (Li et al., 2018). Microplastics have been identified as a significant threat to marine ecosystems; however, rivers play a crucial role in transporting land-derived plastic waste to the ocean, with an estimated 2.8-18.6% of coastal plastic emissions entering the sea via river transport. While river-dominated coasts comprise only 0.87% of global coastlines, they receive a disproportionate 52% of the plastic pollution delivered by rivers, such as the Ohio River, which meets the Mississippi River, flows into the Gulf of Mexico, and then into the Atlantic Ocean (Horton et al. 2024). While comparable data on microplastics in freshwater environments exist, there are fewer studies associated with freshwater compared to marine environments, with less than 4% of microplastic-related studies reportedly focused on freshwater (Li et al., 2018).

While it is understood that microplastics are present in both freshwater ecosystems and within the tissues of organisms, such as those found in the Ohio River, less is understood about the location of microplastics within the body of specific organisms. Microplastics can enter the bodies of aquatic organisms through their feeding habits or by their location in the water column of a freshwater ecosystem, as microplastics have varying densities.

In this study, the fish species channel catfish (*Ictalurus punctatus*) and bluegill (*Lepomis macrochirus*), commonly found in the Ohio River, were evaluated to determine the distribution of microplastics within their organs. Channel catfish are classified as opportunistic feeders and are benthic inhabitants of freshwater systems, meaning their diet can include a wide range of organisms, especially mussels. Consumption of these organisms may result in the bioaccumulation of plastic pollutants within the organs of the predator. As for bluegill species, they tend to rise to surface waters to feed mainly on aquatic insects (Species Profile - Bluegill). The concern arises with this foraging strategy due to the most abundant microplastic polymer type, polyethylene (PE), having a lower density than water, which allows it to float on the surface (Rahmayanti et al. 2022).

This study aims to investigate the quantities of microplastics in three organs of channel catfish and bluegill: the swim bladder, gills, and stomach. More specifically, to determine if feeding habitats and foraging strategies influence the location of microplastics within common fish species in the Ohio River.

METHODS

Fish specimens were collected on an opportunistic basis within the Ohio River via gill net, hoop net, or by electrofishing. After a specimen was obtained, it was placed in a -20° C freezer until dissection. Before dissection, the specimens were thawed completely.

Dissection: Extractions of the swim bladder and stomach were performed by use of Metzenbaum scissors, probe, and a scapula. During dissection, the ventral midline of the fish was cut starting from the anus to the pectoral fin to expose the abdominal cavity. As for the gills, the operculum was lifted and cut at its base to expose and extract the gills adequately.

Digestion: After dissection, all tissue samples were weighed and placed in a 150 mL beaker along with 10 mL of hydrogen peroxide, 10 mL of 0.05 M Fe (II) solution, and 20 mL of 10% KOH. The 0.05 M Fe (II) solution was prepared in bulk by adding 500 mL of distilled water, 3 mL of sulfuric acid, and 7.5 g of sulfate heptahydrate. Each addition to the digestion solution was made slowly, as the mixture tended to foam and rise to the surface in the beaker. All tissue samples were then set to digest on a hot plate set at 65°C for 48 hours with aluminum foil placed over each beaker.

Filtration: After tissue samples were digested, each beaker containing the digested mixture was filtered using a vacuum filtration system, a side-arm flask, qualitative filter paper, glass microfiber filter paper, a Hiersch funnel, and tubes. Each mixture underwent two filtrations. The first filtration used Whatman Grade 1 Qualitative paper to catch the larger particles, allowing the microplastics to filter into the funnel below. The second filtration utilized the previous filtrate and microfiber filter paper to ensure the recovery of microplastics.

Analysis: Each filter paper was set to dry for 24 hours in a vented petri dish. After drying, each filter paper was viewed under a microscope for microplastic quantification and identification. To ensure the validity of the data, each particle that appeared to be a microplastic was evaluated using the hot needle test. In this evaluation, a hot needle is placed in proximity to the questioned particle; if the particle bends or curls, it is deemed a microplastic. If the object shows no reaction to the hot needle, it is determined to be a biological particle from the tissue. The identification of microplastics was evaluated using the Guice to Microplastic Identification of the Marine & Environment Research Institute (MERI). Pictures of microplastics were also sized and analyzed using ImageJ software.

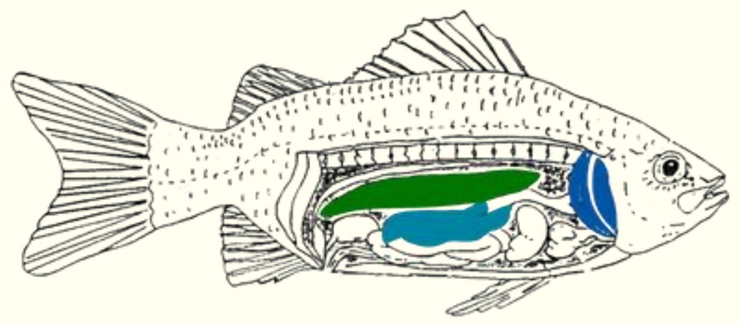

Figure 1: General internal fish anatomy derived from (Fish Internal Anatomy Coloring 2025) that highlights the swim bladder (green), stomach (teal), and gill (blue).

Figure 2: Organs obtained after dissection. The swim bladder is shown to the left (green), stomach is shown in the middle (teal), and gills are shown to the right (blue).

RESULTS

Table 1: Total amount of microplastics per organ per fish species.

Species	Organ	Mass of organ tissue (g)	Amount of microplastics
I. punctatus	Gill tissue	4.368	5
	Stomach tissue	9.228	9
	Swim bladder tissue	8.806	7
I. punctatus	Gill tissue	3.039	2
	Stomach tissue	8.632	4
	Swim bladder tissue	3.585	5
L. macrochirus	Gill tissue	0.085	4
	Stomach tissue	0.485	2
	Swim bladder tissue	0	0
L. macrochirus	Gill tissue	0.141	3
	Stomach tissue	0.508	4
	Swim bladder tissue	0	0

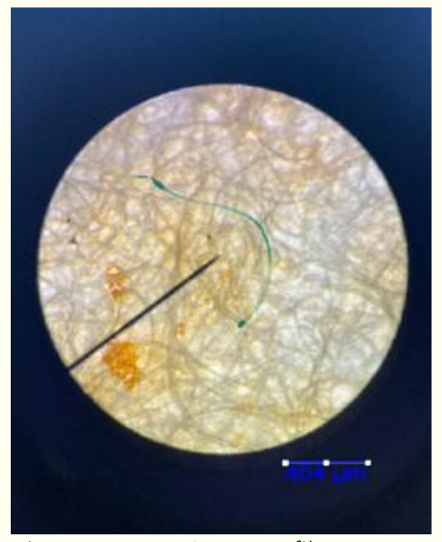

Figure 3: Green filamentous microplastic found in the digested stomach tissue of I. punctatus.

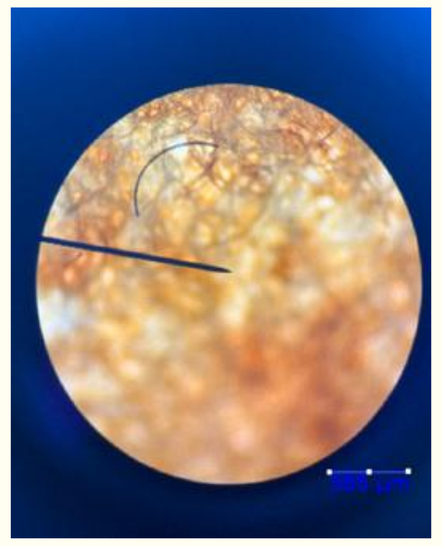

Figure 4: Blue/black filamentous microplastic found in the swim bladder of I. punctatus .

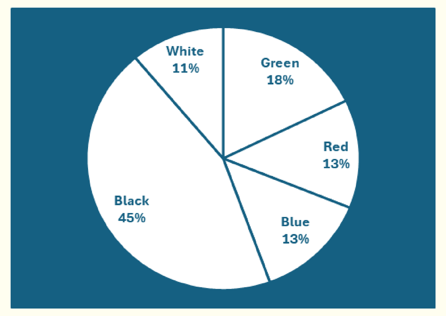

Figure 5: Pie chart compiled of colored microplastic data across all fish specimen.

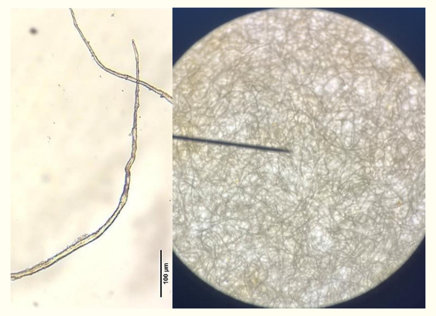

Figure 6: Transparent filamentous microplastic (left) derived from (MERI) and appearance of filter paper under the microscope (right).

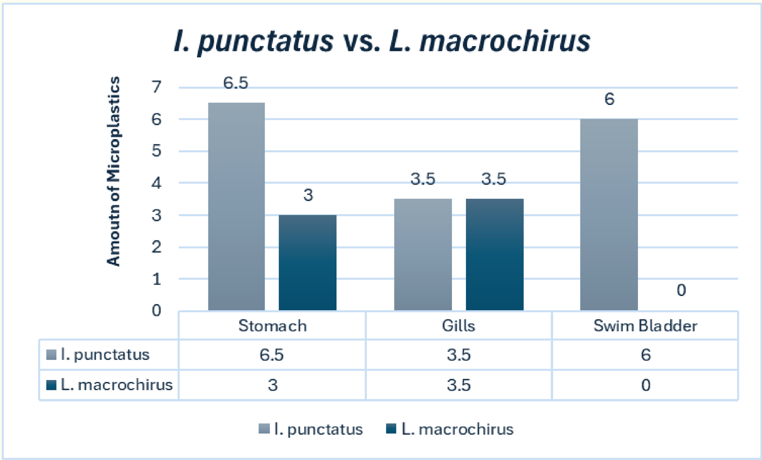

Figure 7: Chart exhibiting the average number of microplastics per organ in both fish species: I. punctatus and L. macrochirus.

Filament microplastics were the most abundant across all samples, especially those to be black filamentous. Transparent filaments were not accounted for in this study due to the filter paper having the same appearance to ensure validity of the data (Figure 6). All sample collections of *L. macrochirus* also had deflated swim bladders, leading to no recovery of microplastics from the organ.

I. punctatus revealed to have the highest amount of microplastics in the stomach and swim bladder (Figure 7). As for the microplastic amount in the gills, both species *L. macrochirus* and *I.punctatus* had the same average amount during analysis.

CONCLUSIONS

Microplastics were detected in all examined organs of both channel catfish (*Ictalurus punctatus*) and bluegill (*Lepomis macrochirus*), with black filamentous fibers dominating in gills, stomach, and swim bladder tissues. Benthic-feeding catfish exhibited the highest amount of microplastic particles in the stomach and the swim bladder tissues, especially due to swim bladder tissue not being retrieved from *L. macrochirus*. This suggests that feeding ecology and habitat use can drive organ-specific accumulation. Whereas pelagic *L. macrochirus* exhibited lower amount of microplastics in the stomach and same amount of microplastics as *I. punctatus* in gill tissue. It is important to note that the bluegill specimen collected were smaller in size compared to the channel catfish specimen collected from the Ohio River. For example, in Table 1, the average mass of the gills in *L. macrochirus* was 0.113g, while the average mass of the gills in *I. punctatus* was 3.704g. While the mass of gills in bluegill collected were significantly lower than those in channel catfish, both species had an average amount of 3.5 microplastics in their gill tissue (Figure 7). This data confirms that the habitat usage of *L. macrochirus*, such as depending on prey near the surface of water, could expose them to microplastics of lower density in through their gills when intaking surface water for oxygen.

Although the sample size was limited, these findings highlight the potential for microplastics to enter freshwater trophic systems and their concern for bioaccumulating in key organs with unknown consequences. By demonstrating that benthic species like *I. punctatus* accrue more particles in the lower gut-tissues while pelagic foragers such as *L. macrochirus* showed the same amount of microplastics in the gills as channel catfish, the results highlight how habitat use mediates contaminant pathways.

Future studies could expand sample sizes across seasons and life stages to validate these trends, employ higher resolution polymer identification methods, and evaluate the direct effect of trophic transfer to predators. Ultimately, integrating organ-level microplastic quantification with ecological and toxicological assessments will be essential for developing mitigation strategies to implement and protect aquatic ecosystem health.

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FURTHER INFORMATION

For further information or questions about this study please contact Lindsay Barnes at Lindsaybarnes0915@gmail.com or lnbarn55@thomasmore.edu.